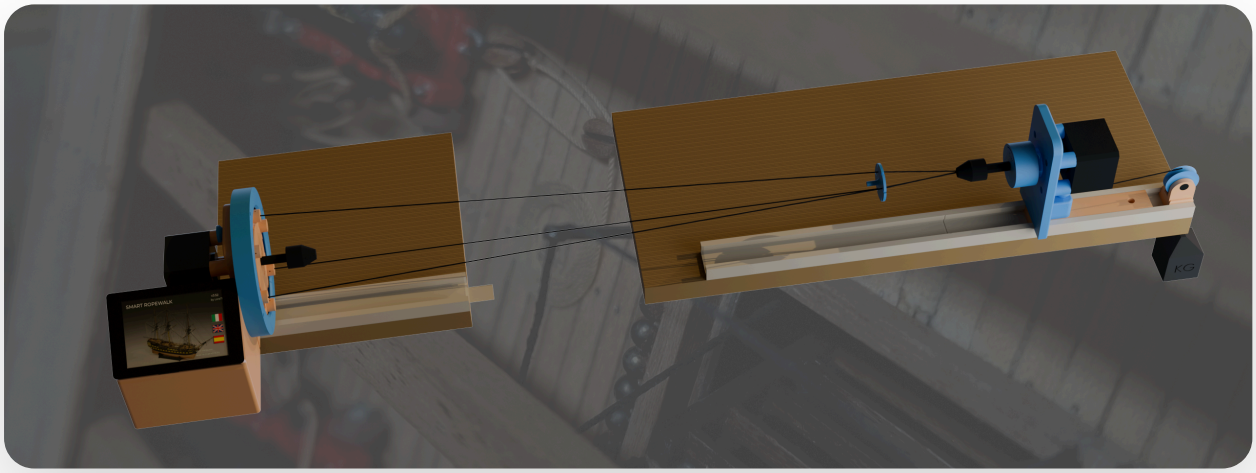
Smart Ropewalk

User guide



English version - firmware v3.66

1. Introduction and activation

Smart Ropewalk is an electronic machine designed to make scale rope and cables for ship modelling.

All functions are controlled from the touchscreen: motor settings, automatic cycle, serving, parameter calculation, recipes, manual controls, Wi-Fi and firmware updates.

This guide describes the screens and controls of the English version.

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Activation and software protection

Smart Ropewalk uses a digital license bound to the display MAC address. The license protects the software from unauthorized duplication and identifies each produced machine correctly.

1 Display MAC

The MAC is shown on the startup screen, at the lower right. This code identifies the display and is used to generate the license.

2 license.dat file

The license is supplied as a **license.dat** file and must be copied to the root of the display microSD card.

3 Demo mode

If the license is missing or does not match the MAC, the display remains accessible but the motors are disabled. When a motor start is requested, the DEMO MODE message is shown.

4 Duplication

A license copied to another display will not enable the machine, because the hardware MAC does not match the authorized one.

Note: keep a copy of the license associated with your display. After the first activation, the firmware also stores an internal backup copy.

2. Main screen

The main screen is the operating centre of the machine. It shows the active recipe, both motor settings, cycle progress and the most frequently used controls.



Main screen

1 Active recipe
Name and summary of the recipe in use: material, strands, length and tension.

2 Serving
Opens the Serving function to cover a rope with fine thread.

3 Wi-Fi status
The icon shows whether the display is connected to Wi-Fi.

4 Twist
Selects the phase and controls for Motor 1, used for twisting.

5 Motor 1 RPM
Touch the value to set Motor 1 speed.

6 Motor 1 direction
Sets CW or CCW for twisting.

7 Wind
Selects the phase and controls for Motor 2, used for winding.

8 Motor 2 RPM
Touch the value to set Motor 2 speed.

9 Motor 2 direction
Sets CW or CCW for winding.

10 Target T/W ratio
Twist-to-winding ratio specified by the recipe.

11 Help
Opens the machine help page.

12 Twists
Current and target twist counter.

13 Windings
Current and target winding counter.

14 Actual ratio
Live T/W ratio during the operation.

15 START
Starts or resumes the selected phase.

16 STOP / PAUSE
Stops motion; in AUTO, retains counters for resuming.

17 RESET
The only command that clears the counters.

18 AUTO
Runs twisting, stops in pause, then prepares winding.

19 Presets
Opens the list of stored recipes.

20 Manual controls
Opens independent controls for both motors.

21 Rope calculation
Opens the recipe calculator and rigging functions.

22 Wi-Fi
Opens Wi-Fi connection, OTA updates and database functions.

Tip: During an automatic cycle, use STOP/PAUSE to interrupt the process temporarily. Counters are reset only by RESET.

3. Rope calculation

The calculator suggests operating parameters from the thread data and the desired result.



Rope calculator screen

1 Desired diameter
Required final rope diameter.

2 Thread diameter
Diameter of the starting single thread.

3 Material
Cotton, linen, wool and silk: same natural-fibre model. Polyester: separate model. E121 120(1): dedicated recipes for an untwisted filament.

4 Total strands
Total number of threads in the rope.

5 Groups
Number of groups wound together.

6 Initial length
Length of the strands before processing.

7 Tension
Shows the counterweight to use to keep the rope under tension.

8 Twists
Calculated number of twists.

9 Windings
Calculated number of windings.

10 Obtainable
Estimated diameter from the selected configuration.

11 Error
Difference from the desired diameter.

12 Thread twist
Shows whether the starting thread is RH, LH or untwisted.

13 Final length
Estimated completed rope length.

14 RH / LH
Selects the final lay: RH rope or LH cable.

15 SHIPS
Opens kit search and lets you select the model and scale.

16 RIGGING
Opens the list of rigging for the selected ship.

17 APPLY
Applies the calculated recipe to the main work page.

18 SAVE
Saves the calculated recipe as a preset.

Indicative tension ranges: up to 0.19 mm 100 g; 0.20-0.50 mm 150 g; 0.51-1.20 mm 250 g; 1.21-1.80 mm 350 g; over 1.80 mm 450 g. For single threads below 0.10 mm and some fine cotton configurations, the calculator applies precautionary exceptions that require experimental verification.
Tested and catalogued materials: 0.22 mm RH cotton; 0.10 mm LH 140/2 cotton; 0.12 mm LH 100/2 cotton; 0.28 mm RH standard polyester; 0.10 mm untwisted E121 120(1) polyester; 0.10 mm untwisted Skala 240; 0.08 mm untwisted Skala 360; 0.34 mm RH Mara 70; 0.50 mm RH Mara 30. The menu also includes 0.09 mm RH 170/2 cotton, which still needs validation. Cotton, linen, wool and silk are natural fibres and use the same empirical cotton model.

4. Serving

This function drives both motors together to wrap a fine thread around a core or an already-made rope.



Serving screen

1 Speed

Sets the rotation speed used during serving.

2 Direction

Selects CW or CCW rotation.

3 Reverse command

Reverses the selected rotation direction.

4 Turn counter

Shows the turns completed in the current operation.

5 Stop target

Number of turns after which the machine stops automatically.

6 Diameter calculation

Wind a known number of rope turns on a smooth, thin cylindrical support. Enter the number of turns and the total measured width: the calculator returns the rope diameter.

7 START

Starts serving.

8 STOP

Stops the operation.

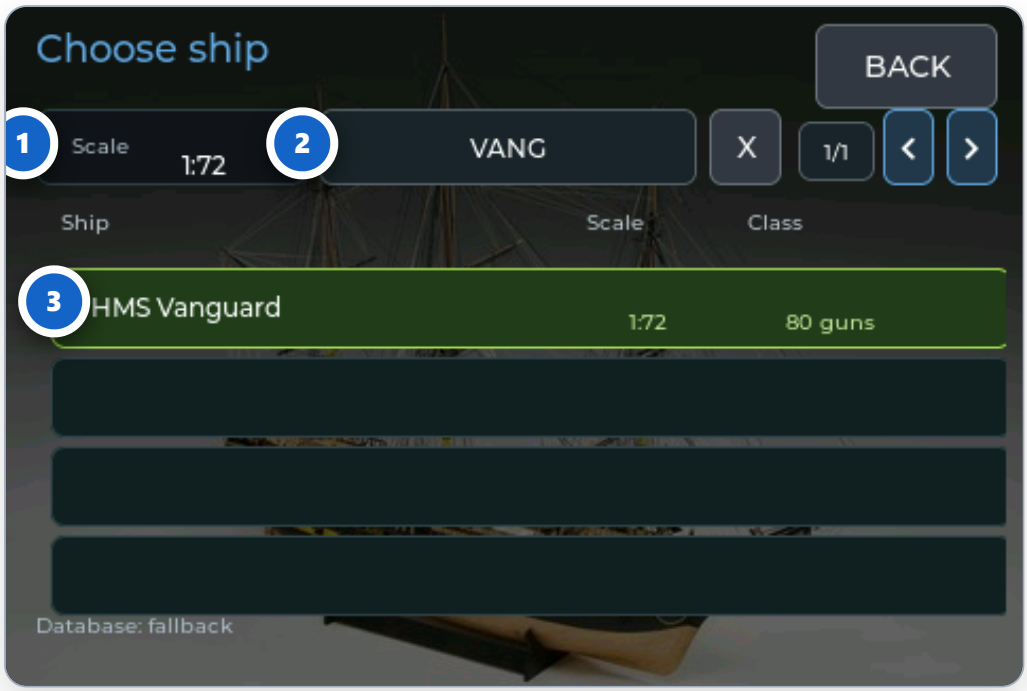
9 RESET

Resets the turn counter.

Tip: Before starting, make sure the serving thread can run freely and that the target matches the length to be covered.

5. Ship selection

The Ships page lets you choose the historical reference vessel and its commercial scale. The list contains commercially available model kits and is not exhaustive. The selection remains stored after restarting.



Ship selection screen

1 Model scale

Enter or modify the model scale. The suggested value is taken from the selected kit.

2 Ship search

Filters the list using any part of the vessel name, including partial words. For example, searching for “BLA PR” finds “Black Prince”.

3 List and confirmation

Touch a row: the selection turns green. Use arrows, scrolling or the page number to browse; press BACK to confirm and load the database. This can take several seconds.

Note: the database of listed ships covers most model kits currently on the market. If a ship can be assigned to a class in the Steel tables, the related rigging items and their corresponding diameters are also available; otherwise, rigging items are not available.

6. Rigging list

After confirming a ship, the Rigging page shows the lines used on that vessel according to its Steel class. If the vessel cannot be classified, the data are unavailable. On first access the firmware reads the database from the microSD and displays LOADING DATABASE: WAIT.



Rigging list screen

1 Ship and search

Shows the active ship and scale. The search field filters the rigging list by keyword, including partial matches.

2 Rigging rows

Real: actual rope diameter. **Scale**: diameter reduced to the selected model scale. **Norm.**: diameter obtainable from available recipes. **R** indicates RH rope; **L** LH cable. **n.d.** means that the historical table provides no usable diameter: the row remains readable but cannot be applied to a recipe. Touch a row for full detail.

3 Pages

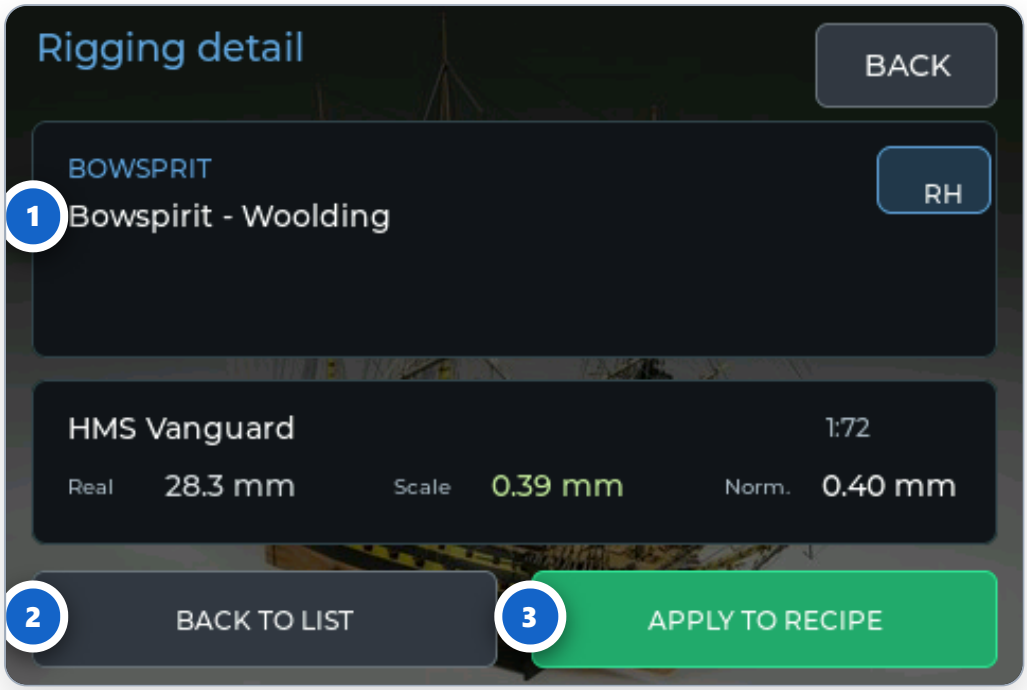
Use arrows, scrolling or the page number to move through the extended list.

4 Database and results

Shows the version of the rigging database in use and the number of rigging items found by the current search criterion.

7. Rigging detail

The detail page shows the full text of the selected rigging item and can send its scaled diameter to the rope calculator.



Rigging detail screen

1 Full description

Shows the complete rigging name without truncation.

2 BACK

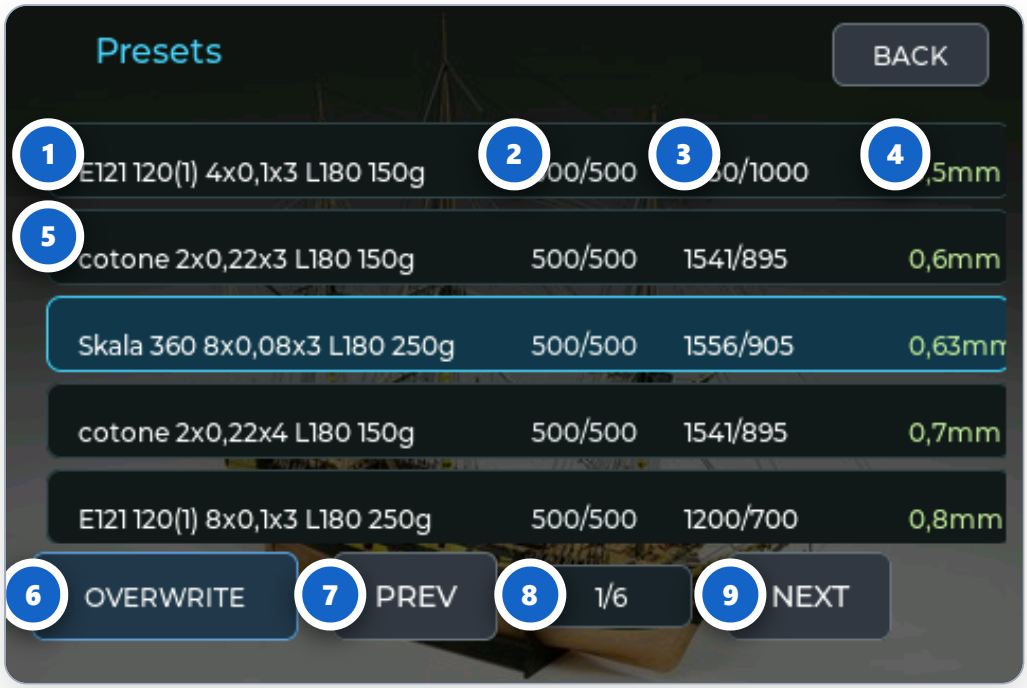
Returns to the rigging list.

3 APPLY TO RECIPE

Sends the scaled diameter to the rope calculator to generate the closest recipe.

8. Presets and recipe editor

The Presets page contains 30 saveable recipes. The first 15 slots contain validated default recipes, but they can be overwritten like every other slot. Recipes saved on the SD card and in internal memory are preserved during firmware updates. Keep a backup of the SD card /recipes/ folder.



Preset screen — recipe list

1 Selected recipe
Touch a row to select it or open its editor.

2 Motor RPM
Speeds set for Motor 1 and Motor 2.

3 Twists / windings
Work targets stored in the recipe.

4 Normalized diameter
Final obtainable diameter specified by the recipe.

5 Other recipes
The rows below contain the other recipes on the page.

6 OVERWRITE
Saves the current recipe in the selected slot, including the 15 default presets.

7 PREV
Moves to the previous page.

8 Page
Shows the current page; touch it to enter a page directly.

9 NEXT
Moves to the next page.

Tip: Touch the desired recipe to select it. Use the Preset editor screen to modify its values.

Recipe editor

The editor lets you review and modify all parameters of the selected recipe, including twists, windings, total strands and groups.



Preset editor screen

1 Recipe name
Automatic or customised preset name.

2 Twist RPM
Motor 1 speed.

3 Wind RPM
Motor 2 speed.

4 Twists
Target number of recipe twists. Touch the value to edit it with the numeric keypad.

5 Windings
Target number of recipe windings. Touch the value to edit it with the numeric keypad.

6 Obtainable diameter
Predicted final rope diameter. Touch the value to recalculate the recipe automatically.

7 Strand diameter
Diameter of each starting thread.

8 Material
Material used for the recipe.

9 Strands
Shows total strands and number of groups. Touching the field asks first for total strands, then for the number of groups.

10 Length
Initial length of the thread.

11 Tension
Weight or pretension applied during the operation.

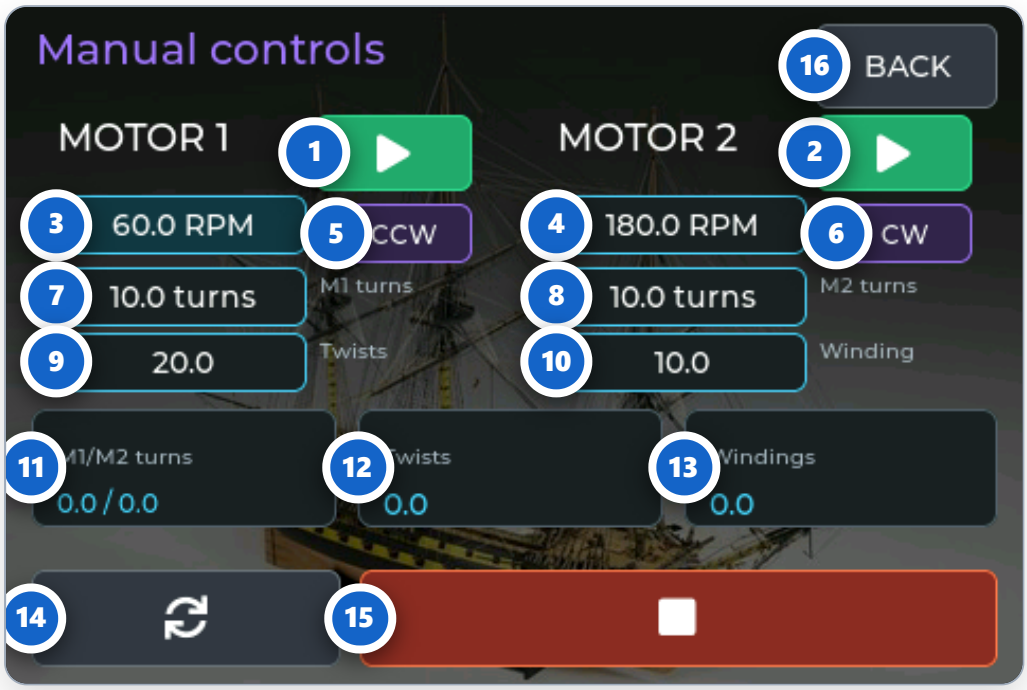
12 APPLY
Applies the recipe to the main screen and saves the current preset data.

13 SAVE
Stores the changes in the selected preset.

Tip: Touching a numeric value opens the numeric keypad. The Strands field requires two consecutive entries: total strands and groups.

9. Manual controls

The Manual controls page lets you test and adjust the two motors independently.



Manual controls screen

1 START / STOP motor 1
Starts or stops motor 1.

2 START / STOP motor 2
Starts or stops motor 2.

3 Motor 1 RPM
Sets the speed of motor 1.

4 Motor 2 RPM
Sets the speed of motor 2.

5 Motor 1 direction
Selects CW or CCW for motor 1.

6 Motor 2 direction
Selects CW or CCW for motor 2.

7 Motor 1 turn target
Sets the turn limit for motor 1.

8 Motor 2 turn target
Sets the turn limit for motor 2.

9 Twist target
Sets the twist target and automatically updates the motor 1 turn target.

10 Winding target
Sets the winding target and automatically updates the motor 2 turn target.

11 Motor 1 / 2 turns
Shows the estimated counters of the two motors.

12 Twists completed
Shows estimated twists: two per motor 1 turn.

13 Windings completed
Shows estimated windings: one per motor 2 turn.

14 RESET counters
Resets the working counters.

15 STOP all
Stops both motors without resetting the counters.

16 BACK
Returns to the main page.

Note: Change targets while the motor is stopped. Zero selects continuous motion. Counters are estimates, not encoder measurements.

Tip: Use this page to experiment with new recipes and parameters.

10. Wi-Fi

The Wi-Fi page connects the machine to the local network and provides access to updates.



Wi-Fi screen

1 UPDATE
Opens the firmware update page.

2 Connection status
Shows whether the machine is connected and its assigned IP address.

3 Network name / SSID
Field for the Wi-Fi network name.

4 Password
Password field; the icon shows or hides it.

5 RESET WI-FI
Clears or resets the stored Wi-Fi data.

6 SCAN
Scans for available networks.

7 SAVE
Saves SSID and password and attempts to connect.

8 CONNECT / DISCONNECT
Connects to or disconnects from Wi-Fi.

Tip: Wi-Fi is only required for firmware updates and any network services.

11. Firmware and database update

The Update page checks online for newer firmware and rigging database releases and lets you check the license activation status. During a firmware update, the three PDF manuals on the microSD are automatically updated if the online package has a newer date or a PDF is missing. A manual download error does not prevent the firmware update.



Firmware update screen

1 OTA status
Current firmware update message.

2 Network and IP
Active Wi-Fi network and display IP address.

3 Network name
Name used to identify the machine on the local network.

4 Installed firmware
Version currently installed on the display.

5 Online firmware
Version available after the online check.

6 Firmware result
Result of the latest firmware check or installation.

7 SEARCH
Checks online for newer firmware and database files.

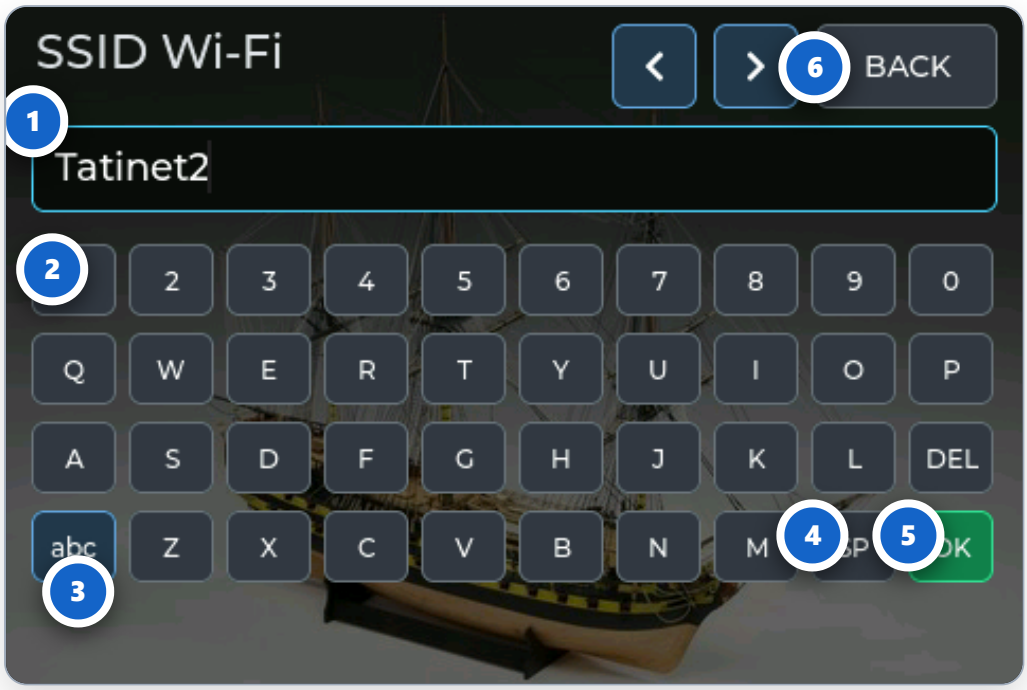
8 Firmware update
Shows or starts the available update; without news it shows NO UPDATE.

9 Database
Shows the result of the rigging database check.

Database and license: DB UPDATE downloads and installs the rigging database when a newer version is available. CHECK LICENSE verifies this display's activation status online. Do not remove power while an update is being installed.

12. Alphanumeric keyboard

The alphanumeric keyboard is used to enter the SSID and password.



Wi-Fi keyboard screen

1 Text field
Shows the entered text.

2 Keyboard
Numeric and alphabetic keys.

3 abc / ABC
Switches between lower-case letters, upper-case letters and symbols.

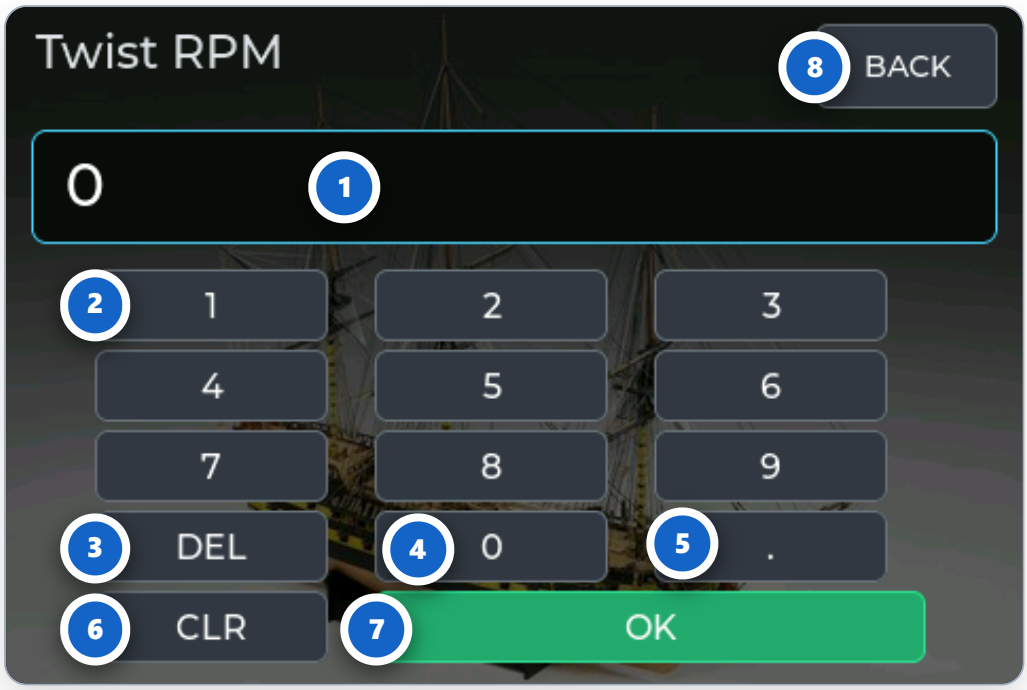
4 SP
Inserts a space.

5 OK
Confirms the text.

6 BACK
Cancels and returns to the previous page.

13. Numeric keypad

The numeric keypad opens when a numeric parameter is touched.



Numeric keypad screen

1 Value field
Shows the value being edited.

2 Digits
Enter the new value.

3 DEL
Deletes the last digit.

4 Zero
Enters 0.

5 Decimal point
Enters a decimal point.

6 CLR
Clears the entire value.

7 OK
Confirms the new value.

8 BACK
Closes without confirming.

14. Help

The built-in help summarises the machine's main functions.



Help screen

1 Main
Explains the main controls and value selection.

2 Serving
Summarises the serving function.

3 Auto
Describes the automatic cycle.

4 Presets
Explains loading, saving and overwriting recipes.

5 Calcolo
Explains the empirical test database and validated recipes.

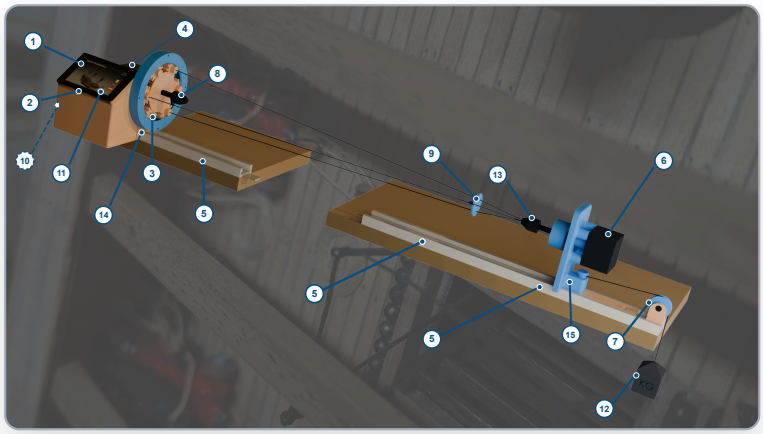
6 Manual controls
Summarises independent use of both motors.

7 RESTART
Restarts the system.

8 BACK
Returns to the previous screen.

15. Machine setup

The diagram identifies the main Smart Ropewalk components and their role in preparing the machine.



1 Touch display
Control interface for recipes, cycles and settings.

2 Electronic unit
Houses the control electronics and connections.

3 Six twisting heads (hooks)
Hold and rotate the strand groups.

4 Motor 1
Drives strand twisting.

5 Linear guide
Supports the Motor 2 assembly; one section can be used to secure Motor 1 to the workbench.

6 Motor 2 and carriage
Drives rope laying.

7 Tensioning pulley
Guides the line to the tensioning weight.

8 Serving spindle 1
Dedicated spindle for rope serving.

9 Strand separator
Keeps strands separated during twisting.

10 Main power switch
Turns the electronic unit on or off.

11 microSD slot
Insert the microSD card before powering on the display.

12 Tensioning weight
Hang the recipe-specified weight from the line routed through the terminal pulley.

13 Spindle 2
Second spindle on the right, connected to Motor 2, where the strands converge.

14 Motor 1 support
Base of the Motor 1 assembly, to be fixed to the work surface.

15 Motor 2 support
Support for the Motor 2 assembly, movable along the linear guide.

Tensioning: hang the weight from the terminal pulley and secure the line to the pin on the carriage, as shown in the diagram. Weights and the line used to suspend them are not included.

Preparation and assembly

The Smart Ropewalk is supplied pre-assembled. Prepare the work surface and check every connection before first use.

Package contents

- 12 V, 6 A power supply.
- Motor 1 unit with support, gears, electronics enclosure and display.
- Motor 2 unit with support, two 3 mm spindles, two four-core motor leads and an extension lead for Motor 2.
- Six 200 mm interlocking guide sections, counterweight pulley with support and strand separator.

Installation

Fix the Motor 1 unit on the left side of the work surface, either directly or by using an interlocking guide. It must not move while working. To use the Serving or Rope diameter calculation functions, place it at the edge of the bench, preferably overhanging it, so the serving thread can descend vertically from the spindle.

Assemble a linear guide of about 800 mm for ropes with an initial length of up to 1.8 m. Join the sections carefully without steps; use a small amount of grease or oil if necessary. Fix the guide to the bench, insert the Motor 2 carriage and place the terminal pulley behind it.

Note: the two assemblies do not necessarily need to be placed on the same table. What matters is that the two spindles are aligned and sufficiently far apart. The distance between the spindles is the length of rope to be made, allowing for shortening caused by twisting. The machine has been tested with a working length of 1.8 m; different lengths will produce proportionally different recipes.

Tensioning weight: secure the line to the pin on the Motor 2 carriage, pass it over the pulley and hang the required weight. Fishing weights of 30 g, 50 g, 70 g and 100 g combinations are suitable. Weights and suspension line are not supplied.

16. Working procedure

Making a rope

From the main screen, press the pen-shaped icon (21) to open the Rope calculation page.



1. Select the desired final rope diameter (1).
2. Select the final lay: RH rope or LH cable (14).
3. Select the initial rope length, namely the distance between the two spindles (6).
4. Select the material from the quick-selection menu (3). For a known material, its thread diameter is entered automatically in the Thread diameter field. For an unlisted material, enter the thread diameter (2) and its original twist: RH, LH or untwisted (12).
5. Fit suitable counterweights following the tension proposed automatically in (7).
6. Prepare the machine by distributing the total number of strands in (4) across the number of hooks/groups in (5).
7. Fit the strand separator near the Motor 2 spindle to prevent groups from tangling during twisting.
8. Press APPLY (17) to transfer the recipe to the main work page, or SAVE (18) to store a preset.

Note: on this page it is also possible to set the number of strands and the number of groups (hooks). The calculator then shows the corresponding final diameter and counterweight, and calculates twists and windings.



1. Pressing APPLY returns to the main work page: the recipe sets the twist direction (6), winding direction (9), required twists (12) and windings (13).
2. Press AUTO (18) to start the machine: Motor 1 performs the twists until the specified count is reached.
3. When the required twists are reached, the machine stops: remove the strand separator.
4. Press PLAY (15): the machine winds the rope until the required winding count is reached, then stops.

Note: at any time, STOP/PAUSE (16) stops the motor and PLAY (15) resumes it. Motor speeds can also be changed with controls (5) and (8).

5. At the end of the process, secure the rope ends with a small drop of cyanoacrylate and remove the rope.
6. Stretch the completed rope under reasonable tension, according to material and diameter, without breaking it.

Note: it is normal for the completed rope to unwind by several turns before settling, especially with synthetic threads such as polyester.

Directions: for RH rope, Motor 1 = CCW and Motor 2 = CW. For LH cable, Motor 1 = CW and Motor 2 = CCW. The recipe sets the directions automatically, but they remain editable.

17. Permitted use and limits

The machine is an experimental project for model makers. It has been tested with an initial length up to 1.8 m and a final rope diameter up to 2 mm. Above 2.5-3 mm, tension and mechanical loads increase considerably: avoid these configurations to prevent damage to 3D-printed parts.

Do not exceed 500 RPM: above this value the motors may stop or lose synchronisation.

Firmware and database can be updated over Wi-Fi. Recipes are based on experimental work and may need adjustment for the material in use.

Thank you for your confidence,



Lorenzo Lotti

18. Web Simulator

The Web Simulator lets you operate Smart Ropewalk from a computer using a larger screen. It shows the same pages as the display and sends clicks and touches to the machine: calculations, counters and movements remain controlled by the firmware. The machine must be switched on and connected; the panel is not a standalone offline simulator.

Connecting

1. Connect the machine and computer to the same Wi-Fi or local network.
2. Download the Web Simulator ZIP attached to the firmware release and extract the entire archive onto your computer.
3. Keep **index.html**, **styles.css**, **protocol.js**, **simulator.js** and the three files **readme_EN.txt**, **readme_IT.txt**, **readme_ES.txt** together.
4. Open **index.html** in your browser. No application installation is required.
5. Find the machine's IP address in the display's Wi-Fi information, enter it in **Display address**, then click **Connect**. If the browser requests permission to access the local network, allow it.

Using the panel

Click or tap the controls just as you would on the physical display. All pages, menus, keyboards and popups are available. Drag within lists to scroll, or use the arrows. Changes made on the physical display also appear in the browser at the next refresh.

The surrounding panel controls are in English; machine screens follow the language selected on the display. The firmware version is shown below the screen, and the interface follows the installed release as long as it supports the panel.

Refresh selects an interval of 1, 2 or 5 seconds; the default is 2 seconds. The circular arrow button refreshes the screen immediately. Web animations are less smooth than on the physical display; counters show actual data at the time of capture. Requests pause while the browser tab is hidden.

Activation: the Web Simulator does not bypass the licence. In demo mode the pages remain available, but motors cannot be started, just as on the physical display.

Warning: closing the browser, clicking **Disconnect** or losing Wi-Fi **does not stop the motors**. If the connection is lost, use the machine's physical touchscreen to stop them. When the database is loading, wait for completion without repeating commands.